PULSE-HMS: Complete Role-by-Role Hospital Workflow

Who logs in first, what each role does, and how to explain the patient lifecycle to an interviewer.

System Clean Slate Status:
All old test records have been reset. Beds are 100% Available (Green). Patients count is 0.
You will add everything with your own hands by following the 6 stages below!

The 6-Stage Hospital Lifecycle (Role-Handoff)

STAGE 1: FRONT DESK INTAKE

1. Log in as RECEPTIONIST

Login with: Username: `receptionist` | Password: `Password@123`

What the receptionist does in a hospital: Registers the patient and books the doctor appointment.

1. **Register Patient:**
   - Click **Patients** in the sidebar (`/patients`).
   - Click **+ Register Patient**.
   - Enter your own name: First Name: **Sumanth**, Last Name: **Puli**, Gender: **Male**, DOB: **1996-05-15**, Phone: **+91 98765 43210**, Blood: **O+**, Emergency Contact: **Pooja (+91 98765 43211)**.
   - Click **Save Patient Record**. A green badge appears with your code: `PAT-2026-0001`.

2. **Book Doctor Appointment:**
   - Click **Appointments** (`/appointments`) → **+ Book Appointment**.
   - Select Patient: **Sumanth Puli**, Dept: **Cardiology**, Doctor: **Dr. Rajesh Varma**, Time: **Tomorrow 10:00 AM**, Reason: *"Chest discomfort and routine checkup"*.
   - Click **Confirm Appointment**.
   - Change status dropdown to `CONFIRMED` (patient has arrived in the waiting lounge).

→ *Now click your Profile icon (top-right) and click **Sign Out**.*

STAGE 2: DOCTOR CONSULTATION ROOM

2. Log in as DOCTOR

Login with: Username: `dr.rajesh` | Password: `Password@123`

What the doctor does in a hospital: Conducts clinical examination, records ICD-10 diagnoses, logs vitals, and prescribes medicine.

1. **Consultation Queue:**
   - Go to **Appointments**. See **Sumanth Puli** waiting. Change status to `IN_CONSULTATION`.
2. **Clinical EMR Chart & Vitals:**
   - Go to **Clinical EMR** (`/emr`). Select **Sumanth Puli** at the top.
   - Click **+ New Encounter**.
   - Enter Complaint: *"Patient reports exertion fatigue and morning dizziness."*
   - Enter Assessment & Plan: *"Mild hypertension. Prescribe Amoxicillin. Follow-up in 10 days."*
   - Enter Vitals: BP **125/82**, Pulse **74**, SpO2 **98%**.
   - Click **Save Encounter Record**. His chart updates with ICD-10 diagnosis `I10 Essential Hypertension`.
3. **Write Electronic Prescription:**
   - Go to **Prescriptions** (`/prescriptions`) → **+ New Prescription**.

- Patient: **Sumanth Puli**, Doctor: **Dr. Rajesh Varma**.
- Add Medicine: **Amoxicillin 500mg** (1 tablet, Twice daily, 5 days).
- Click **Save Prescription**.

→ *Click Profile icon → **Sign Out**.*

STAGE 3: HOSPITAL PHARMACY COUNTER

3. Log in as PHARMACIST

Login with: Username: `pharmacist` | Password: `Password@123`

What the pharmacist does in a hospital: Fulfills doctor prescriptions, inspects batch expiry dates (FEFO), and adjusts stock inventory.

1. Go to **Pharmacy Inventory** (`/pharmacy`).
2. Inspect the active medication catalog and batch `AMO26001` (Amoxicillin 500mg).
3. Click **Adjust Stock** on batch `AMO26001`:
   - Adjustment Quantity: `-10`
   - Reason / Type: `Return to Vendor` (or Damaged)
   - Audit Notes: *"Inspected packaging before dispensing"*
   - Click **Record Adjustment**.
4. Verify: Stock deducts smoothly in real-time, and an audit trail entry is logged under the **Transactions** tab!

→ *Click Profile icon → **Sign Out**.*

STAGE 4: INPATIENT WARD ADMISSION

4. Log in as STAFF NURSE

Login with: Username: `nurse` | Password: `Password@123`

What the nurse does in a hospital: Checks the visual Bed Board and admits the patient to an available ICU or General Ward bed for observation.

1. Go to **Inpatient Bed Board** (`/bed-board`).
2. Review the 5 visual bed cards. Notice all 5 beds are green (`AVAILABLE`).
3. Click **Admit Patient** (or click Bed `B-101-A`):
   - Patient: **Sumanth Puli**

- Admitting Physician: **Dr. Rajesh Varma**
- Bed: **Bed B-101-A - Room ICU-101 ($250/day)**
- Reason: *"Overnight observation and cardiac telemetry"*
- Click **Confirm Admission**.

4. Verify: Bed `B-101-A` immediately turns RED (`OCCUPIED`) displaying Sumanth Puli's badge and admission ID!

→ *Click Profile icon → **Sign Out**.*

STAGE 5: CASHIER & BILLING DESK

5. Log in as ACCOUNTANT

Login with: Username: `accountant` | Password: `Password@123`

What the accountant does in a hospital: Compiles all hospital fees into an invoice, collects payments, and issues financial receipts.

1. Go to **Billing & Payments** (`/billing`).
2. Click **+ Generate Invoice**.
3. Select Patient: **Sumanth Puli**.
4. Add Billable Charges:
   - Consultation Fee: **$120.00**
   - ICU Daily Bed Rate: **$250.00**
   - Pharmacy Medications: **$30.00**
5. Click **Create Invoice**. The total is $400.00 with status `UNPAID`.
6. Click **Record Payment**:
   - Amount: **$400.00** | Method: **Credit Card**
   - Click **Submit Payment**.
7. Verify: Status changes to `PAID` and Remaining Balance updates to $0.00!

→ *Click Profile icon →* ***Sign Out****.*

STAGE 6: HOSPITAL EXECUTIVE OVERSIGHT

6. Log in as SUPER ADMIN

Login with: Username: `superadmin` | Password: `Password@123`

What the director does in a hospital: Reviews real-time census, department workload, bed occupancy, and system health.

1. Go to **Dashboard** (`/dashboard`).
2. Notice your live numbers:
   - Total Registered Patients: **1**
   - Occupied Beds: **1 / 5 (20% Occupancy)**
   - Today's Appointments: **1**
3. Hover over the **Weekly Consultation Trends** and **Department Workload** charts to inspect the clean tooltip cards!
4. Open Swagger Docs: `http://localhost:8085/swagger-ui/index.html` to show 40+ REST APIs.

How to Explain this Flow to the Interviewer

> *"Our system mirrors the real-world healthcare workflow across 6 distinct hospital roles:*
>
> *1. **Receptionist** registers the patient and schedules the consultation.*
> *2. **Doctor** conducts the clinical exam, logs ICD-10 diagnoses in the EMR, and prescribes medication.*
> *3. **Pharmacist** inspects batch stock and fulfills medicines using FEFO (First-Expiring, First-Out).*
> *4. **Nurse** manages the Bed Board and admits the patient to an available ICU bed with concurrency locks.*
> *5. **Accountant** compiles consultation, bed, and drug charges into an invoice and records payment.*

> *6. **Administrator** monitors live analytics, census KPIs, and security audit logs across the hospital."*